# DEVRAJ PARAJULI | @devpro13 @devpro13 devrajparajuli.com.np

+977-9861333460 Jawalakhel, Lalitpur, Nepal devrajparajuli13@gmail.com

## EDUCATION

- **Bachelor in Engineering** | ***Electronics and Communication*** | ***GPA: 3.22*** Nov. 2019 – April 2024
  **Tribhuwan University, Institute Of Engineering, Pulchowk Campus, Pulchowk, Nepal**
  **Major subjects**: Basic-Advanced Electronics, Microprocessor, C++, Embedded Systems, Control Systems, Filter Design, Digital Signal Analysis and Processing,Computer Architecture, Electromagnetics, Data Structure, Digital Logics, Communication Systems,Telecommunication Systems, Antenna, RF microwave and AI etc.

**Major Interests: VLSI Design**, **Hardware Accelerators In-Memory Computing**, **Secure Designs**, **Microelectronics Systems**, **CAD Tools**.

## SKILLS

- **Programming:** Programming fundamental and OOP Concepts, C++, C, JAVA, Python, SystemVerilog(Design+Validation), RUST
- **Hardware:** RISC-V Architecture based processor design and testing(Learning), 8085/8086 MPU, 8051/52 microcontroller, Assembly, KICAD designing, PCB fabrication, drilling, soldering, Arduino, MCU-peripheral interfacing.
- **Tools and libs:**Git, LateX, Proteus, Modelsim
- **ML tools and Libs:**Numpy, Pandas, PyTorch and Sklearn(Learning)
- **Personal Skills:**Teamwork, Collaboration, Research and Problem Solving

## PUBLICATIONS

- S. K. Khatri, **D. Parajuli**, P. M. Mane, B. Chaulagain, and S. Thapa, "**Fall Detection System for Elderly People Using LSTM**", J. Eng. Iss. Solut., vol. 5, no. 1, pp. 210–220, Apr. 2026.
  This study addresses the consequences and effects of falls on elderly people and provides a low-powered wearable hardware solution embedded with an LSTM model for detecting fall events to enhance elderly safety.

## PROJECTS

- **Design and Verification of a Fixed-Point Attention Accelerator** | ***SystemVerilog/Python*** 2026
  Designed and verified a SystemVerilog RTL implementation of the Transformer self-attention mechanism using fixed-point arithmetic, lookup tables for exponential calculations, and parallel MAC units. Validated hardware outputs against a Python reference model to evaluate accuracy.
- **Design of GNN Accelerator** | ***SystemVerilog*** 2026
  Studied GNN architectures and explored RTL implementation of core computational stages, including graph aggregation, feature multiplication, and state-machine-based control using SystemVerilog.
- **Internet of Medical Things and Streaming** | ***C++/Python*** 2025
  Contributed in sensors and MQTT protocol configuration; Developed Interactive Dashboards.
- **RISC-V-Complete-Multicycle-Processor Design** | ***SystemVerilog*** 2024
  Modeled RISC-V complete multicycle processor and designed using SystemVerilog.
- **Design of Star Tracker Simulator for the Attitude Determination of Spacecrafts** 2024
  Part of **undergraduate research major project** where I developed of Image Simulator Software; inegrated hardware components with the PI processor.
- **Fall Detection System** | ***Python, Hardware System*** 2023
  Part of **undergraduate research minor project** where I designed the circuitry for the wearable device in KICAD and fabricated it in PCB

- **Dijkstra's Algorithm Visualizer** | *C++* 2022
  Implemented the concept of Data Structure like Nodes, linked list, Stack, Tree, Map etc in this program.
- **OCR Using OpenCV and Google's Tesseract API** | *C++* 2021
  A small tool for optical character recognition; Contributed in making environment setup, and application development;Implemented OpenCV in C++.

## Work Experience

- **Part time Lecturer** Apr 2025–Present
  **Tribhuvan University**, Lalitpur, Nepal
  - Lectured undergraduate Electronics Engineering courses in Microprocessors, Computer Organization and Architecture and Embedded Systems.
  - Conducted practical laboratory sessions involving digital circuit design, microcontroller programming and hardware interfacing.
  - Guided students in practical implementations, debugging, and project-based learning activities.
- **SDET / Software Engineer - AI Automation** June 2024–Present
  **Cedar Gate Technologies an IQVIA Business**,Sanepa Lalitpur, Nepal
  - Worked on SDLC processes, Cloud Computing and developed enterprise product knowledge base for GenAI applications by defining business rules and designing LLM skills.
  - Enhanced interpersonal skills, documentations skills and learned about professional ethics.
- **Teaching Assistant** Aug 2024-Nov 2024
  **IOE, Pulchowk Campus**
  - Mentored students in core Python programming concepts for the Samsung Innovation Campus (SIC), focusing on problem-solving, code optimization, and clean coding practices
  - Facilitated hands-on learning through supervised projects and coding sessions, while assisting instructors in content delivery and interactive session management to ensure effective learning outcomes.
- **Undergraduate Research Fellow** | ***IOE Pulchowk Campus*** Nov 2022-May 2024
  - Conducted undergraduate research on reliable fall detection systems using inertial sensor data and machine learning techniques.
  - Investigated LSTM-based approaches for detecting and classifying human falls from time-series IMU data.
  - Worked on sensor data processing, algorithm development, and experimental validation for research prototypes.
  - Contributed to the design of image simulator for star tracker for attitude determination applications.
  - Presented research findings and contributed to the technical documentation and development of research projects.
- **Hardware Coordinator** | ***Robotics Club, IOE Pulchowk Campus*** 2023-2024
  - Coordinated hardware events on Campus; Actively participated in ABU Robocon Competition.